Teacher answers and topic guidance

Accept equivalent science in the usual response mode. Keep answers hidden during independent attempts.

Starter

1. Cell division.
2. Differentiation.
3. In the root-tip meristem.

Repair an evidence claim

A longer root supports growth. Length alone cannot establish cell identity or the relative contribution of division and elongation. The blood-cell drawing models a possible lineage for the named adult source. It cannot prove a treatment outcome or a wider cell range.

Build the growth chain

Division increases number. Elongation increases length. Differentiation gives a specialised role. For the fresh skin case, more cells support division. Look for features or function before naming a specialised type.

Scope

Embryonic stem cells can give rise to many body-cell types. Adult blood-forming stem cells have a narrower range. Cells in root-tip meristems divide. In skin repair, new cells can replace damaged cells.

Arrival Supported

1. Stem cell.
2. More cells.
3. The nucleus.
4. Root tip.

Arrival Independent

1. Stem cell.
2. More cells with matching inherited information.
3. Division makes more cells; differentiation gives a specialised role.
4. In the root-tip meristem.

Arrival Aspire

1. Stem cell.
2. Inherited DNA is copied; two daughter cells receive matching inherited information.
3. Division is shown. Differentiation explains specialised roles.

4. The new cells can elongate, increasing their length.

Exit Supported

1. Division.

2. The root grew.

Exit Independent

1. Division increases cell number.

2. Evidence of a characteristic feature or function.

Exit Aspire

1. Cell count shows a change in number. Differentiation requires evidence about specialised features or roles.

2. It models a possible blood-cell lineage. Treatment success would require outcome evidence.

Made by Matt

Mark scheme Supported

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Match division, elongation and differentiation to more cells, longer cells and a specialised job. [3]

Division adds cells. Elongation lengthens cells. Differentiation gives cells features suited to a job.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Compare damaged skin and a growing root. Name one shared process and one feature of the root route. [2]

Both use division. In the root, division happens in the meristem or new cells can elongate.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

3. A root is longer. Choose what the evidence supports: the root grew, or every root cell is a stem cell. Explain your choice. [2]

The root grew. Length alone does not identify the types of cells in it.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Independent

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Explain how division and differentiation can help replace damaged skin cells. [3]

Division adds cells. Some descendants differentiate and acquire features suited to their roles. New cells replace damaged cells.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Compare skin repair and root growth. Use division, meristem and elongation. [3]

Both use division to add cells. Root-tip meristem cells divide. New root cells can elongate, contributing to length.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

3. Repair this claim: more cells in a tissue proves that new specialised cell types formed. Use the cell count and name one extra check. [3]

The count supports an increase in cell number. It does not identify specialised types. Compare features with a labelled reference or gather evidence of their function.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Aspire

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. A root is longer and the model shows longer cells. Explain how division and elongation could contribute. State what length alone cannot establish. [3]

Division can add cells. Elongation increases cell length. Root length alone does not tell which process contributed or identify all cell types.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Repair: an adult blood-forming stem cell can become any body cell. Use the blood-cell example and compare it with an embryonic source. [3]

Adult blood-forming stem cells give rise to blood-cell types. Embryonic stem cells can give rise to many body-cell types. The adult example does not support the claim about any body cell.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

3. A model labels every new skin cell as a stem cell. Correct two labels or claims and justify your edits with number and role evidence. [4]

Use division for the increase in cell number and justify it using the count. Identify specialised descendants only when there is evidence of features or a role and link this to differentiation.

Maximum 4 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Choice

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Choose division or differentiation for more cells. Choose elongation or division for longer cells. Give one reason. [3]

Division adds cells. Elongation increases length. A reason distinguishes the two changes.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Choose both, skin only or root only for making more cells by division. Name one root feature. [2]

Both. Accept a root-tip meristem or elongation of new root cells.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

3. A root is longer. Choose growth or proof of stem-cell identity as the supported claim. Give a reason. [2]

Growth. Length does not tell us the identity of each cell.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt